Persistent Topological Divergence for Provable Hallucination Detection

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The Hallucination Problem

What LLMs Do Well

- Generate fluent text
- Answer complex questions
- Perform reasoning tasks

The Critical Flaw

- Generate plausible but **false** information
- Create internally consistent **fabrications**
- Undermine trust in high-stakes domains

How do we detect when LLMs hallucinate?

Existing Detection Methods

Paradigm	Core Idea	Key Limitation
Knowledge-Grounded	Check facts against external databases	Latency, coverage, cost
Self-Consistency	Generate multiple outputs, check agreement	High computational cost
Model-Internal	Analyze hidden states	Requires training data
structure.fg!10 **TOHA (Topological)**	Measure connectivity to prompt	**Misses complex patterns**

Problem: Existing topological methods only measure *connectivity*. They miss higher-order structural inconsistencies.

What TOHA Misses

TOHA's approach: Measure how "connected" response tokens are to the prompt

Failure Case: "Connected but Incoherent"

A hallucinated response can:

- Have strong token-by-token attention to the prompt
- Form an internally consistent but **factually incorrect** narrative loop
- Score **low** on TOHA (looks well-connected)

Why this happens:

- TOHA only uses H_0 (connectivity, connected components)
- Blind to H_1 (loops), H_2 (voids), higher structures
- Measures minimum spanning forest length with linear error in response length

Persistent **To**pological **Div**ergence for **E**valuation

Key Idea: Dual Topological Views

A faithful response must be:

1. **Internally coherent** (has its own structure)
2. **Grounded to the prompt** (anchored to context)

Hallucination = Large mismatch between these two views

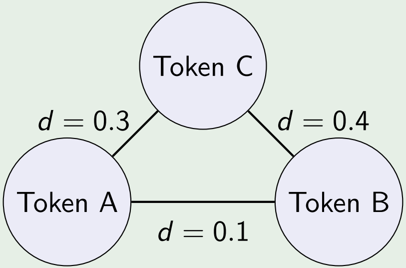
From Attention to Topology

Step 1: LLM produces attention matrix $W = [w_{ij}]$ where $w_{ij} \in [0, 1]$

Step 2: Convert to pseudo-distances: $d(i, j) = 1 - w_{ij}$

- Strong attention ($w_{ij} \approx 1$) $\rightarrow$ small distance ($d \approx 0$)
- Tokens are "topologically close"

Example



Persistent Homology: The Key Tool

Idea: Track how topology changes as we vary a scale parameter

Vietoris-Rips Filtration

At scale ϵ, connect tokens i and j if $d(i,j) \leq \epsilon$

As ϵ increases: isolated points $\rightarrow$ clusters $\rightarrow$ loops $\rightarrow$ higher structures

$\epsilon = 0$ 4 components	$\epsilon = 0.5$ 2 components	$\epsilon = 1.0$ 1 component

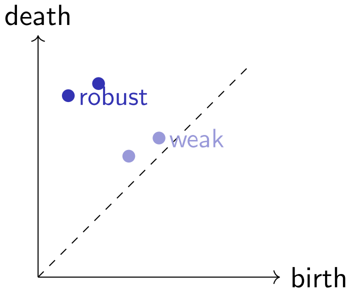
Output: Persistence diagram = birth/death times of topological features

What They Encode

Each point (b, d) represents:

- A topological feature
- Born at scale b
- Dies at scale d
- Persistence $= d - b$

Points far from diagonal = robust features



We compare diagrams using **Wasserstein distance**: optimal matching cost between features

The Two Topologies

1. Internal Response Topology

Question: What is the response's own narrative structure?

Construction:

- Take only response tokens V_R
- Use attention weights between response tokens: W_{RR}
- Build Vietoris-Rips filtration $\rightarrow$ persistence diagram $\mathrm{PD}_k(G_R^{\text{internal}})$

2. Grounded Response Topology

Question: How is each part anchored to the prompt?

Construction:

- For each response token r, define grounding: $f_{\text{ground}}(r) = \min_{p \in V_P}(1 - w_{rp})$
- Small value = strongly connected to prompt
- Build sublevel set filtration $\rightarrow$ persistence diagram $\mathrm{PD}_k(G_R^{\text{grounded}})$

The PerToDiVE Score

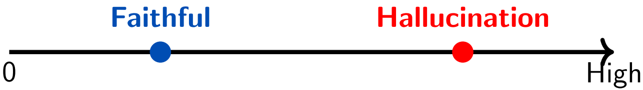
Formal Definition

$$\boxed{\mathrm{PerToDiVE}_k(P, R) = W_p\left(\mathrm{PD}_k(G_R^{\text{internal}}), \mathrm{PD}_k(G_R^{\text{grounded}})\right)}$$

where W_p is the p-Wasserstein distance (typically $p = 2$)

Interpretation:

- **Low score:** Internal structure matches grounding $\rightarrow$ likely faithful
- **High score:** Large mismatch $\rightarrow$ potential hallucination



Visual Example: Internal vs Grounded

Faithful Response

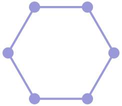
Internal: linear

Grounded: linear

Similar diagrams

Low PerToDiVE

Hallucinated Response



Internal: loop (H_1)

Grounded: disconnected

Very different!

High PerToDiVE

Theoretical Guarantees: Stability

Question: Is PerToDiVE robust to small numerical perturbations?

Lipschitz Stability Theorem

If attention weights change by at most δ, then:

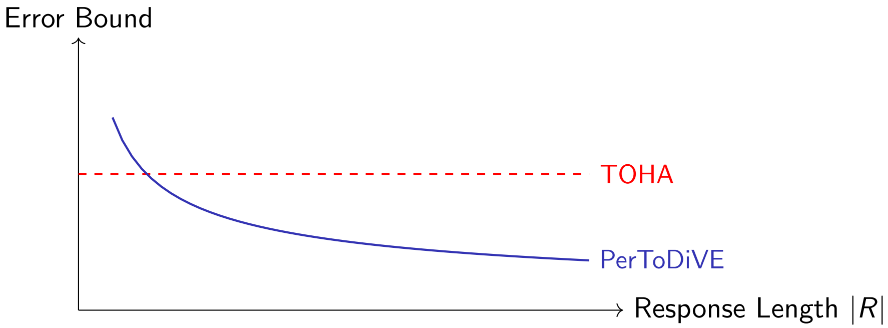
$$|\Delta \text{PerToDiVE}_k| \leq C_k \cdot \delta$$

where $C_k = |V_R|^{1/p}$ for the p-Wasserstein distance.

Method	**Raw Error Bound**	**Per-Token Error**
TOHA	$\lvert R\rvert \cdot \delta$	δ (constant)
structure.fg!10 **PerToDiVE**	$\lvert R\rvert^{1/p} \cdot \delta$	$\lvert R\rvert^{1/p-1} \cdot \delta$

Key insight: For $p > 1$, our error *decreases* as responses get longer!

Why Sub-Linear Scaling Matters



For long, complex generations

PerToDiVE is provably more reliable for long-form text

Critical for detailed explanations, reports, and reasoning

Higher-Dimensional Features Matter

TOHA limitation: Only uses H_0 (connectivity) **PerToDiVE advantage:** Captures H_k for all dimensions

What Different Dimensions Detect

- H_0: Connected components H_1: Loops (circular reasoning)
- H_2: Voids (gaps in reasoning) Higher H_k: Multi-scale inconsistencies

Example: Circular Reasoning

A hallucinated response creates a self-reinforcing loop of claims grounded in each other but not the prompt:

- High H_1 persistence in internal topology
- Low H_1 persistence in grounded topology
- Large Wasserstein distance $\rightarrow$ detected!

Formalizing "Ungrounded" Responses

Definition: (k, τ)-Topologically Ungrounded

A response R is (k, τ)-ungrounded if:

- Its internal topology has significant features with persistence $\geq \tau$
- Its grounded topology has mostly features with persistence $< \tau$

Proposition

If R is (k, τ)-ungrounded for large τ, then $\text{PerToDiVE}_k(P, R)$ is large.

Intuition: The Wasserstein distance must pay a high cost to match persistent internal features to weak/absent grounded features.

This formalizes the connection: structural mismatch $\Rightarrow$ high score $\Rightarrow$ hallucination signal

Computational Complexity

Algorithm overview:

1. Extract attention matrix W from LLM
2. Compute internal persistence diagram (VR filtration on V_R)
3. Compute grounded persistence diagram (sublevel filtration)
4. Calculate Wasserstein distance between diagrams

Complexity Analysis

- Persistence computation: $O(n^3)$ worst case, faster in practice
- Wasserstein distance: Near-linear with kd-tree auction algorithm
- **Total:** Practical for real-time detection

Single forward pass required (unlike self-consistency methods). Can be computed for selected attention heads only.

Advantages Over Prior Methods

Why PerToDiVE is Better

vs. TOHA: Captures higher-dimensional features ($H_1, H_2, \ldots$), not just connectivity; sub-linear error scaling instead of linear

vs. Self-Consistency: Single forward pass (not multiple generations); no sampling overhead

vs. Knowledge-Grounded: No external database required; works for proprietary or specialized domains

Provable Guarantee: First multi-scale topological detector with formal Lipschitz stability theorem

Comparison Table

Property	TOHA	SelfCheck	Knowledge	PerToDiVE
External data	No	No	Yes	No
Multiple passes	No	Yes	Varies	No
Training data	No	No	Yes	No
Dimensions	H_0 only	N/A	N/A	All H_k
Stability bound	Linear	None	None	Sub-linear
Detects loops	No	Limited	N/A	Yes
Formal proof	Weak	None	None	**Strong**

PerToDiVE combines the best properties: efficient, model-internal, and provably robust

Key Contributions

Dual-Filtration Framework

Compare internal vs grounded structure

Multi-Dim. Topology

Capture loops, voids, not just connectivity

Provable Stability

Sub-linear error scaling with length

First topological detector with formal multi-scale guarantees

Future Directions

Immediate Next Steps

- **Empirical validation:** Test on benchmark datasets (TruthfulQA, HaluEval)
- **Optimize computation:** Implement efficient persistence algorithms
- **Head selection:** Develop principled methods for identifying informative attention heads

Longer-Term Research

- Extend to other model architectures (non-Transformer models)
- Explore connections to causal interventions in attention
- Develop real-time monitoring systems for deployed LLMs
- Investigate topological signatures of different hallucination types

Broader Impact

Scientific

- Bridges TDA and NLP communities
- Opens new research directions in interpretable AI
- Provides toolkit for analyzing attention mechanisms

Practical

- Enables safer LLM deployment
- Applicable to high-stakes domains (medicine, law)
- Framework-agnostic approach

Moving from *heuristic detection* to *provable reliability*

The PerToDiVE Framework

- **Problem:** LLM hallucinations undermine reliability; existing detectors are incomplete
- **Solution:** Compare internal response topology with prompt-grounded topology using persistent homology
- **Innovation:** First multi-dimensional topological detector with provable sub-linear stability
- **Impact:** Moves field toward principled, theoretically grounded detection methods

Open Questions

Theoretical

- Can we tighten the stability bound further?
- What is the information-theoretic limit of topology-based detection?
- How do different persistence dimensions correlate with hallucination types?

Practical

- Which attention heads are most informative across different models?
- How does performance scale to very long contexts (100K+ tokens)?
- Can we adapt this framework to multimodal models?

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